Contact

www.linkedin.com/in/vijaydaultani (LinkedIn)
vijaydaultani.github.io/ (Personal)

Top Skills

Machine Learning
Convolution Neural Networks
Natural Language Processing (NLP)

Languages

Japanese (Limited Working)
English (Full Professional)
Hindi (Native or Bilingual)
Sindhi (Native or Bilingual)

Certifications

Sequence Models
Microsoft Certified Professional (MCP)
Deep Learning Specialization
1Z0-007 Introduction to Oracle9i:SQL
Machine Learning

Publications

Sparse Direct Convolutional Neural Network (ISNN 2017)
Acceptability Evaluation of Naturally Written Sentences
Improving Automatic Evaluation of Acceptability Based on Language Models with a Coarse Sentence Representation
Unsupervised Extractive Summarization for Product Description using Coverage Maximization with Attribute Concept.
The Importance of Weather for E-Commerce Orders Forecasting

Patents

Vijay Daultani

CEO & Co-Founder @ Neural Nurture | Ex-Amazon, Rakuten, NEC | PhD @ Tokyo Tech | MTech @ IIT Delhi | Visiting Scholar @ UC Berkeley
Indore, Madhya Pradesh, India

Summary

I am a Research Scientist with a PhD in Natural Language Processing from Tokyo Institute of Technology, specialising in linguistic-acceptability modelling. My journey spans both deep academic research and hands-on industry leadership in machine learning, NLP and applied AI.

At Rakuten (2021-2023) I led the NLP Group and steered cross-domain initiatives in E-Commerce, Finance and Travel — including keyword ad-bidding optimisation and knowledge-graph based item-attribute extraction. Prior to that, at Amazon Japan (2019-2021) I served as Senior Technical Program Manager in Applied ML, managing global projects across 13 countries and building core NLP infrastructure (e.g., Japanese-tokeniser, sequence-to-sequence ad-generation pipeline).

My early career included research roles at NEC Central Research Laboratories (2014-2017) where I developed deep-learning algorithms that achieved up to 95 % speed-ups on CNNs for supercomputers, as well as a visiting scholar stint at UC Berkeley's Swarm Lab focusing on IoT and vertical mobility.

With a foundation in a Computer Science degree from TCET Indore and an M.Tech from IIT Delhi (CGRA compilers under Prof. Anshul Kumar), I bring together high-performance computing, deep learning and NLP to build research-driven, production-ready solutions.

I'm passionate about translating academic insight into meaningful business impact, mentoring ML talent, and advancing AI systems that scale. I'm always exploring collaboration opportunities, new challenges at the intersection of research and industry, and ways to leverage intelligent systems for real-world problems.

Information processing method and device for neural network.

Information processing apparatus, information processing method and storage medium storing program

Optimising computational and memory efficiency of CNN.

P.S. Check out my website at https://vijaydaultani.github.io/

Experience

Neural Nurture
CEO & Co-Founder
May 2025 - Present (1 year 6 months)
Indore, Madhya Pradesh, India

• Leading the vision, research, and development of cutting-edge AI solutions in NLP and Computer Vision at Neural Nurture (www.ntwo.ai)
• Building strategic partnerships and a strong multidisciplinary team to drive impactful innovation and real-world deployment of AI technologies

Rakuten
Group Manager
October 2021 - July 2023 (1 year 10 months)
Tokyo, Japan

• Led a global team of 15 scientists to develop cutting-edge NLP solutions across E-Commerce, Finance, and Travel domains, enhancing project delivery and cross-functional collaboration.
• Directed a keyword advertising initiative focused on optimizing bidding strategies for external search engines, driving improved campaign performance.
• Designed the framework for a Knowledge Graph project aimed at extracting and organizing product attributes for Rakuten’s E-Commerce platform, with a focus on improving data quality and future scalability.
• Contributed to the development of Rakuten’s in-house Japanese large language model (Rakuten AI 7B LLM), supporting advanced AI capabilities for business applications.

Amazon
Sr. Technical Program Manager - Applied ML
August 2019 - September 2021 (2 years 2 months)
Greater Tokyo Area

• Directed a cross-functional team of 7 scientists to design and deploy advanced NLP solutions.
• Make suitable tradeoffs to optimize time-to-market and clearly communicating goals, roles, responsibilities and desired outcomes across cross-functional teams

• Scope complex projects and deliver simple, elegant solutions to challenging problems with significant customer and revenue high-impact business problems

Automated Advertisement:
• Co-invented new machine learning (sequence to sequence) algorithms to generate search queries for automated advertisement in paid search channels
• Interacted with a broad cross section of the Amazon organization to clarify ambiguous issues and negotiate effective technical solutions between development and business teams, to ensure on-time product delivery
• Demonstrated communication and interpersonal skills including relationship building and collaboration with a cross-function team across 13 countries

Japanese Tokenizer:
• Co-authored internal research paper on novel metrics and to measure impact of segmentation on downstream tasks of information retrieval.
• Co-authored internal research paper on novel neural network-based lexicon-free Japanese text segmenter. Delivered integration of the segmenter in 6 business use cases

Rakuten
2 years 6 months

Head of NLP (Natural Language Processing) Team
July 2018 - July 2019 (1 year 1 month)
Tokyo, Tokyo, Japan

Fraud Detection:
• Created deep learning (FFNN, RNN) based fraud detection platform for predicting fraudulent credit card transactions for Rakuten Card. Discovered optimal transaction sequence representation for RNN.
• Headed deep learning based hybrid platform utilising text and images features for predicting fraudulent recipes for Rakuten Recipe

Text Summarization
• Invented new unsupervised algorithm for text summarization, filed a patent & published research paper
• Utilized technology for summarizing financial reports, blogs and news articles to assist investment analysts to make quick decisions for Rakuten Investment Management
• Utilized technology to experiment and explore automatic generation of meeting minutes for Rakuten Mobile

Machine Translation:

• Improving neural machine translation model for Rakuten.com (global E-Commerce platform) for translating item pages for cross border trading in 4 languages (Japanese, English, Korean, Chinese)

Search Query Prediction:

• Invented new supervised method for automatic generation of user search queries and filed a patent

• Automatically generated user search queries were used to make models to deliver better user experience and product recommendations

Sentiment Analysis:

• Led project to perform Recurrent Neural Network based sentiment analysis for the reviews across platforms of Rakuten.

Attribute Extraction & Normalization:

• Led project to extract item tags (attributes) and normalizing entities (values) using semi-supervised machine learning algorithms such as CRF and RNN for Rakuten Ichiba

Product Recommendations:

• Initiated and led project to extract user preferences from images for better product recommendations

AI Trainer

• Conduct AI Trainings for Technology Division and Businesses.

Deep Learning Researcher

February 2017 - June 2018 (1 year 5 months)

Tokyo, Japan

• Predicting user search queries for E-commerce Business.

• Predicting segment of retarget-able users for Travel Business.

NEC Corporation

Researcher

October 2014 - February 2017 (2 years 5 months)

Kawasaki-ku, Kanagawa, Japan

• Experimenting deep learning with convolutional neural network on NEC's SX architecture, inventing ways for optimal hardware utilisation for acceleration.

• Invented a new technique for optimal instruction selection of a compiler, filed a patent and initiated a project for it.
• Profiled, analysed bio metrics application, modified data structures and proposed algorithms to suite the underlying hardware architecture.
• Understand different optimisation phases of the state of the art compilers like icc, gcc and llvm. Understand the problems and find their solutions which when provided to the compilers they can generate code more suitable for the hardware architecture.

University of California, Berkeley
Visiting Scholar
November 2015 - December 2015 (2 months)
San Francisco Bay Area

• Researcher for collaborative project between NEC and UCB. The project was related to giving a optimal advice to the users whether to choose the conventional elevator system or stair case or escalator, by calculating and comparing the service time for each transportation medium respectively.
• Initiated, designed and executed as a lead developer, an IOT project with swarm lab of UCB.
• Integrating several technologies like python, Java and Android, a complete prototyping project was accomplished in one month.

Indian Institute of Technology, Delhi
Teaching Assistant
July 2012 - July 2014 (2 years 1 month)
Greater Delhi Area

• Assisting faculty members with research and classroom activities and grading papers.
• Advising and supervising undergraduate students for their assignments and projects. Showing them the direction so that they can focus their energy and time on learning wonderful and core concepts of the assignments.

Accenture
Associate Software Engineer
September 2010 - August 2011 (1 year)
Mumbai Metropolitan Region

• Excellent learning experience for me; gave me a perception of a tester how he thinks when he tries to find small and big test cases which might have skipped from the developers mind while he was developing the application.

• Now while designing, and developing I keep in mind how to make the software resilient to attacks and also motivates me to prioritise tasks in my bug finding strategies.

Education

Tokyo Institute of Technology

Doctor of Philosophy - PhD, Natural Language Processing · (April 2021 - April 2025)

Indian Institute of Technology, Delhi

Master of Technology (M.Tech.), Computer Science · (2012 - 2014)

T.C.E.T Indore

Bachelor of Technology (B.Tech.), Computer Science · (2006 - 2010)